MIL-S-15253B
29 MAY 1959
SUPERSEDING
MIL-S-15253A
14 JANUARY 1952

MILITARY SPECIFICATION

SCREEN, X-RAY PROTECTIVE

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers sectional and mobile X-ray protective screens for protecting operators from primary and secondary radiation when operating X-ray and photofluorographic equipment.

1.2 Classification. The X-ray protective screens shall be of the following types, as specified (see 6.1):

Type I — X-ray protective screen, sectional, with reusable field type shipping case.

Type II — X-ray protective screen, mobile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

C-F-206 — Felt, Sheet, Wool, Pressed.

FF-C-82 — Casters, Swivel, Chair.

MM-L-751 — Lumber and Timber, Softwood.

QQ-A-267 — Aluminum Alloy 24S; Bars, Rods, and Shapes, Extruded.

QQ-A-268 — Aluminum Alloy 24S; Bars, Rods, and Wire-Rolled or Drawn.

QQ-A-270 — Aluminum Alloy 61S; Bars, Rods, and Shapes, Extruded.

QQ-A-274 — Aluminum Alloy Bars, Rods, and Structural and Special Shaped Sections, Extruded 6063 (63S).

QQ-A-318 — Aluminum Alloy Plate and Sheet 5052.

QQ-L-201 — Lead; Sheet.

QQ-M-31 — Magnesium Alloy, Bars, Rods, and Special Shaped Sections, Extruded.

FSC 6525

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:55Z
Check the source to verify that this is the current version before use.

QQ-S-636 — Steel; Carbon (Low-Carbon), Sheets and Strips.

QQ-W-423 — Wire, Steel, Corrosion-Resisting.

TT-E-527 — Enamel, Synthetic, Lustreless.

TT-E-529 — Enamel, Synthetic, Semi-Gloss.

UU-P-271 — Paper, Wrapping, Waterproofed Kraft.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

MILITARY

JAN-C-496 — Clips, End.

MIL-W-530 — Webbing, Textile, Cotton, General Purpose, Natural or in Colors.

MIL-B-543 — Buckles, Tongueless and Web Strap.

MIL-P-6889 — Primer; Zinc-Chromate, for Aircraft Use.

MIL-S-7952 — Steel, Sheet and Strip, Uncoated, Carbon (1020 and 1025) (Aircraft Quality).

STANDARDS

AIR FORCE-NAVY AERONAUTICAL

AN530 — Screw, Tapping, Thread Forming or Thread Cutting, Round Head, Spaced Thread.

AN545 — Screw-Wood, Round Head.

AN960 — Washer, Flat.

FEDERAL

Fed. Test Method Std. No. 151 — Metals; Test Methods.

Fed. Std. No. 595 — Colors.

Fed. Test Method Std. No. 601 — Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MS20450 — Rivet, Tubular.

DRAWINGS

MILITARY MEDICAL SUPPLY AGENCY

1144 — Mobile Screen, X-Ray Protective.

7006 — Screen, X-Ray Protective-Sectional (3 sheets).

5031 — Shipping Case for Screen, X-Ray Protective (Sectional) (7 sheets) and Bill of Materials.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Satisfactory high grade materials shall be used throughout for both type I, sectional (with reusable field type shipping case), and type II mobile X-ray protective screens. All parts entering into the construction of the protective screens covered by the specification shall be manufactured from materials which are entirely satisfactory for the purpose and of sufficient strength to maintain operational alignment for the life of the equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-18T05:55Z
Check the source to verify that this is the current version before use.

3.1.1 *Type I, sectional.* All material used in the fabrication of the item shall be as designated on Drawing 7006, and shall conform to the applicable specifications listed in section 2 and as specified herein.

3.1.1.1 *Lead sheet.* Lead sheet used shall conform to grade B of Specification QQ-L-201. Lead sheet shall be free of pinholes. The X-ray protective screen shall provide protection against X-radiation over the entire area of the screen of not less than the equivalent of 0.6 mm. of lead at 100 peak kilovolts (PKV) when tested in accordance with 4.4.5.1 and 4.4.5.2.

3.1.1.2 *Lead glass.* Lead glass shall be first class throughout and shall have a lead equivalent of not less than 0.6 mm. at 100 PKV. Glass shall be transparent, clear, and free from flaws, and shall meet the requirements of the lead glass test specified in 4.4.4.

3.1.1.3 *Sponge rubber.* Sponge rubber shall be of a quality that will not assume a permanent set and shall have a durometer reading of not more than 10 and not less than 5 (see 4.4.2).

3.1.2 *Reusable field type shipping case.* All materials used in the fabrication of the shipping case shall be as designated on Drawing 5031 and Bill of Materials thereto, shall conform to the applicable specifications listed in section 2, and as specified herein.

3.1.3 *Type II, mobile.* All materials used in the fabrication of the item covered by this specification shall be as designated on Drawing 1144, shall conform to the requirements of the applicable specifications listed in section 2, and as specified herein.

3.1.3.1 *Lead sheet.* Lead sheet used shall conform to Specification QQ-L-201, shall be grade B, 1.5 ± 0.06 mm. thick, and shall meet the requirements of the X-ray test specified in 4.4.5.1 and 4.4.5.2.

3.1.3.2 *Sheet steel.* Sheet steel used for encasing the lead shall be 0.031 inch (22 gage) if a continuously curved panel is used, or 0.050 inch (18 gage) if a flat panel is used.

3.1.3.3 *Lead glass.* Lead glass used shall be first class throughout, shall have a lead equivalent of not less than 1.5 mm. at 100 PKV and shall meet the requirements of the lead glass test specified in 4.4.4.

3.1.3.4 *Casters.* Casters shall conform to Specification FF-C-82 for type I, size B, material 3. A flat top caster may be substituted providing other requirements are met.

3.2 **Construction.**

3.2.1 *Type I, sectional.* The type I, sectional X-ray protective screen shall be constructed as shown on Drawing 7006.

3.2.2 *Reusable field type shipping case.* The shipping case shall be constructed as shown on Drawing 5031, and all parts shall be as indicated on Drawing 5031 and Bill of Materials thereto.

3.2.3 *Type II, mobile.*

3.2.3.1 *Design.* The design of the screen shall conform generally to Drawing 1144. Where permissible, tolerances are shown. The center of gravity of the screen shall not be higher than 38 inches from the floor. The type II X-ray protective screen shall meet the mechanical requirements specified in 4.4.6.1 and 4.4.6.2.

3.2.3.2 *Window.* The lead glass window shall overlap the lead of the screen by at least ¼ inch, or the window frame shall be of such design and composition as to give equivalent protection. The lead glass shall be supported by a frame which shall hold the glass firmly and close to the screen panel. A rubber channel shall be used to

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:55Z
Check the source to verify that this is the current version before use.

cushion the lead glass. The frame shall be of such material and so designed that the protection provided from X-rays by the glass, frame, and screen panel shall be not less than 1.5 mm. lead equivalent at 100 PKV, with X-rays directed from any angle to any point of the frame. The frame shall meet the requirements of the window frame test specified in 4.4.3. The lead glass window shall be indelibly marked to show the lead equivalent and the voltage at which the equivalent was determined. The lead glass shall be easily removable from the screen.

3.2.3.3 *Base.* The base shall be designed so that the screen panel section can easily be removed from the base. The base shall extend to within ½ inch of the floor, and shall have smoothly rounded corners and edges. The base may be cast or plate.

3.2.3.4 *Screen panel.* The screen panel shall be constructed of the specified sheet lead sandwiched between two layers of the specified sheet steel. The lead shall be of one piece. Joints of any kind are not acceptable. The lead shall be held firmly to the sheet steel by cementing or compression. The vertical edges shall be firmly supported by inserting in a slit tube or rolling around a steel rod. The tube or rod shall be firmly attached to the base. A formed steel or cast cap shall be provided to cover the top edge of the screen panel.

3.3 Finish.

3.3.1 *Type I, sectional.* The entire screen shall be finished with acid-, alkali-, and solvent-resisting enamel, as follows:

3.3.1.1 *Degreasing.* All surfaces to be enameled shall be thoroughly cleaned and chemically degreased.

3.3.1.2 *Primer.* After degreasing, a primer coat shall be applied. The primer shall be zinc-chromate conforming to Specification MIL–P–6889. The primer coat shall form a grounding bond and base for the finishing enamel.

3.3.1.3 *Enamel.* The primer coat shall be followed by two coats of semi-gloss enamel conforming to class B of Specification TT–E–529. Each coat shall be sprayed on and baked in high temperature ovens in accordance with the paint manufacturer's recommendations. The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, and cracking.

3.3.1.4 *Color.* The color of the final finish shall be semi-gloss green conforming to color 24087, as shown in Standard 595 (see 6.2).

3.3.1.5 *Clear enamel.* A final coat of weather-resisting, clear, synthetic baking enamel shall be applied and properly baked.

3.3.2 *Reusable field type shipping case.* The entire unit, including outside and all accessible inside surfaces, shall be finished with acid-, alkali-, and solvent-resisting enamel, as follows:

3.3.2.1 *Degreasing.* All surfaces to be enameled shall be thoroughly cleaned and chemically degreased.

3.3.2.2 *Primer.* After degreasing, a primer coat shall be applied. The primer shall be zinc-chromate conforming to Specification MIL–P–6889. The primer coat shall form a grounding bond and base for the finishing enamel.

3.3.2.3 *Enamel.* The primer coat shall be followed by one coat of lustreless enamel conforming to Specification TT–E–527. Each coat shall be sprayed on and baked in high temperature ovens in accordance with the paint manufacturer's recommendations. The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, and cracking.

3.3.2.4 *Color.* The color of the final finish

Source: https://assist.dla.mil -- Downloaded: 202610-18T05:55Z
Check the source to verify that this is the currentversion before use.

shall be olive drab conforming to color 34087, as shown in Standard 595 (see 6.2).

3.3.3 *Type II, mobile.* The screen shall be finished in a smooth enamel or lacquer which may be in the manufacturer's standard color. The metal shall be properly smoothed, degreased, and primed prior to application of enamel or lacquer. Crackle or wrinkle finishes are not acceptable.

3.4 **Identification marking.**

3.4.1 *Screen.* The front of each screen shall be permanently and legibly marked with the Federal stock number, manufacturer's name or registered trade-mark, serial number, data indicating the amount of protection provided in terms of lead equivalent at 100 PKV, contract number, and the letters "U.S.". Marking shall be accomplished by stamping or by means of a metal nameplate permanently affixed, and shall be of a size such as required for ready readability and identification.

3.4.2 *Shipping case.* Each shipping case, provided with the type I screen, shall be permanently marked on the outside, front and rear, with the Federal stock number, item identification, and the letters "U.S." of a size for ready readability and identification. The inside cover of the shipping case shall be permanently marked showing the manufacturer's name and registered trade-mark. All markings shall be enamel conforming to Specification TT-E-527. Color shall be black, number 37038, as shown in Standard 595 (see 6.2).

3.5 **Instructions.** Instructions including drawings or photographs illustrating assembly and repacking procedures shall be included.

3.6 **Workmanship.** Workmanship shall be first class in every respect. All components shall be properly fitted to assure ease of assembly and disassembly All external surfaces shall be particularly free of sharp edges due to shearing or other manufacturing operations. The screens and shipping case shall be free from defects which detract from their appearance or may impair their serviceability

4. QUALITY ASSURANCE PROVISIONS

4.1 **Supplier responsibilities for inspection.** Such examinations and tests as are set forth in this specification or as shall otherwise be appropriate or necessary to insure that the supplies conform to specification requirements, shall be performed by and at the expense of the supplier. Suppliers who do not have facilities adequate for such tests shall arrange for the use of test facilities acceptable to the Government. Records of examinations and tests performed by the supplier shall be maintained by the supplier and made available to the Government, upon the Government's request, at any time, or from time to time, for a period of 2 years after delivery of the supplies to which such records relate.

4.1.2 Inspection, as used in this specification, is defined as both examination (such as, visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.** Sampling shall be conducted in accordance with Standard MIL-STD-105 and Appendix thereto.

Source: https://assist.dla.mil -- Downloaded: 202610-18T05:55Z
Check the source to verify that this is the currentversion before use.

	Inspection level	AQL (percent defective)
For visual examination		
Major A	II	1.0
Minor A	II	2.5
Major B	II	1.0
Minor B	II	2.5
For dimensional examination		
Screens	L5	2.5
Reusable shipping container	L5	2.5
For component and raw material examination (see 4.1.3)	L5	*
For end item testing (each test)	L5	*

* Acceptance number shall be zero (0) for all sample sizes.

4.2.1 *Raw materials.* Four (4) ounces of each lot of each metal used in the construction of the screens and the reusable shipping case (provided with type I screen) shall be selected for chemical analysis.

4.2.2 *Components.* Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the screens and the reusable shipping case to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated above. No test result shall fall below the minimum or exceed the maximum values specified. The sample unit for testing shall be the quantity necessary to perform all required tests one time each.

4.2.3 Where possible, the same sample unit shall be used for the determination of two or more tests.

4.3 **Examination.** The screens and reusable shipping case shall be examined to determine compliance with all requirements contained in this specification. Nonconformance with these requirements will be permitted to the extent indicated in 4.2.

4.3.1 *Classification of defects.* Inspection shall be conducted in accordance with the following classifications of defects:

Categories and defects **

Major A (Unit of product — Screen)

101. Screen not complete with all specified component parts.

102. Edges and corners of component part not free of burs and sharpness.

103. Metal parts not free of cracks, tears, ruptures, ripples or noticeably deformed areas.

104. Component parts not properly fitted and assembled.

105. Lead glass not transparent, clear and free of cracks and flaws.

106. Spotwelds (on type I screen) not secure and free of cracks and discoloration.

107. Screen does not properly fit into reusable shipping case (Type I Screen).

Minor A (Unit of product — Screen)

201. Rivets in type I screen not securely and properly driven.

202. Fasteners other than rivets not securely tightened.

203. Color not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:55Z
Check the source to verify that this is the current version before use.

204. Identification markings not present or not as specified.

205. Finish not free of bare, chipped out, or excessively scratched areas.

Minor B (Unit of product—Reusable shipping case)

151. Shipping case not complete with all specified component parts.

152. Component parts not free of cracks, tears, ruptures, or noticeably deformed areas.

153. Exposed edges and corners of accessible component parts not free of burs or sharpness.

Major B (Unit of product—Reusable shipping case)

251. Rivets not securely and properly driven.

252. Fasteners, other than rivets, not securely tightened.

253. Identification marking not present or not as specified.

254. Finish not free of bare, chipped out, or excessively scratched areas.

255. Color not as specified.

** Inspection is not restricted to the classified possible defects listed above.

4.3.2 *Dimensional examination.* The screens and reusable shipping case shall be inspected for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 **Tests.** Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Chemical composition.* Tests for chemical analysis shall be conducted in accordance with method 111 or 112 of Fed. Test Method Std. No. 151 (see 4.1.3).

4.4.2 *Sponge rubber.* Test for sponge rubber shall be conducted in accordance with method 3021 of Fed. Test Method Std. No. 601.

4.4.3 *Window frame test.* A 14- by 17-inch film shall be taken with the X-ray film placed behind the screen so that the complete window frame will be projected on the film by an X-ray source placed on the other side of the screen. The film shall be placed parallel to the lead glass window, but the central X-ray beam may be at any angle and directed to any point on the film. The exposure shall be made at 100 PKV, and shall be sufficient to produce a darkening of the film through the lead glass window. Any X-ray leaks, or any evidence of lead tearing, sagging, or flowing at the window corners or in any other area of the screen, shall be considered cause for rejection.

4.4.4 *Lead glass test.* The window shall be removed. A suitable penetrometer shall be placed near one side of the lead glass window and a 14- by 17-inch radiograph taken at 100 PKV. The X-ray exposure shall be examined for comparison purposes. The screen shall be rejected if the lead equivalent is less than 0.6 mm. of lead for the type I screen or less than 1.5 mm. of lead for the type II screen.

4.4.5 *X-ray tests.*

4.4.5.1 With the window removed, a suitable penetrometer shall be placed such that there is an unobstructed path for X-ray through the window opening and that part of the screen near the window can be X-rayed at the same time on one 14- by 17-inch X-ray radiograph. An X-ray radiograph shall be taken in any random area around the window at 100 PKV. The X-ray exposure shall be examined for comparison purposes. The screen shall be rejected if the lead equivalent is less than 0.6 mm. of lead for the type I screen or less than 1.5 mm. of lead for the type II screen.

4.4.5.2 The type I and type II screens shall be tested by X-ray exposure at 100 PKV in at least two random places on each panel of the screens. The 14- by 17-inch X-ray radiographs shall show essentially uniform density.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:55Z
Check the source to verify that this is the current version before use.

4.4.6 *Mechanical tests for type II, mobile.* Two mechanical tests shall be performed as described in 4.4.6.1 and 4.4.6.2.

4.4.6.1 *Rigidity.* With the screen in the normal position, and the base held firmly, a force of 50 pounds is to be applied at any point and in any direction at the top of the screen. Play in any direction shall not exceed 1/8 inch.

4.4.6.2 *Rolling test.* The screen shall be placed on a smooth horizontal concrete floor. Casters shall be clean. A horizontal force shall be applied gently to the top of the screen at any point and in any direction. The force shall be gradually increased until the screen rolls or upsets. If the screen upsets, it is not acceptable.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.* One (1) screen, as specified, constitutes one unit.

5.1.2 *Level A.* Applicable to Stock No. 6525-612-8025 only. Each unit shall be protected by wrapping with kraft paper and cushioning with excelsior pads or flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by tying with cotton cord or tape, or by sealing with kraft gummed paper tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit.

5.1.3 *Level C.* Applicable to Stock No. 6525-612-8025 only. Units shall be protected by standard commercial methods which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.* Applicable to Stock No. 6525-612-8025 only. Each unit shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with Specification PPP-B-601, table II, using type III, class 1 plywood; or PPP-B-621, table IV. Closure and strapping shall comply with appendix of applicable specification. All box panels, except base, shall be lined with waterproof barrier material conforming to the requirements of Specification UU-P-271, class L-2(a), L-2 (b), M-1, M-2, or M-3, in such manner as to prevent direct entry of water. As an alternate, each unit shall be protected with a shroud fabricated of material conforming to Specification UU-P-271, class M-1, M-2, or M-3. The unit shall be adequately secured within the exterior container, by means of bolts, wood braces, blocks, or other suitable devices, to insure against shifting and breakage, or marring of surfaces. The work shall be accomplished in such a manner that the weight of the unit will not be carried directly by its legs or casters.

5.2.2 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Shipping container (see 3.3.2 or 5.2.1) shall be marked in accordance with Standard MIL-STD-129, except that 6.2(c), 6.2(d), 6.5, and 7.5.1, shall not be applicable. Service color and service symbol markings are required only on direct shipments from contractor's plant to overseas activities. For Stock No. 6525-612-8032, the required information shall be printed or stenciled on water-resistant shipping tags which shall be securely affixed to the container by means of metal ties.

6. NOTES

6.1 Ordering data. Procurement documents

Source: https://assist.dla.mil -- Downloaded: 202610-18T05:55Z
Check the source to verify that this is the currentversion before use.

should specify the following:

(a) Number and date of this specification.

(b) Type required (see 1.2).

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 **Availability of color chips.** The individual 3- by 5-inch chips of the colors referenced in 3.3.1.4, 3.3.2.4, and 3.4.2, are available from General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S.W., Washington 25, D.C. Price 5 cents.

6.3 This specification covers the following items appearing in the Armed Services Medical Stock List:

6525–612–8025	SCREEN, X–RAY PROTECTIVE.
6525–612–8032	SCREEN, X-RAY PROTECTIVE.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Army Medical Service
Navy—Bureau of Medicine and Surgery
Air Force

Preparing activity:
Navy—Bureau of Medicine and Surgery (MMSA)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:55Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T05:55Z
Check the source to verify that this is the current version before use.